

Precision Agriculture: A Survey of Techniques

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Abstract—Precision Agriculture (PA) has emerged as a revolutionary mode in the present era of farming, aiming for higher yields with minimum environmental impact. This paper discusses various aspects of PA, underlining its importance in addressing key agricultural concerns, especially resource optimization and sustainability. Advanced techniques such as Variable Rate Technology (VRT), Hydroponics (Vertical Farming), and automation in systems are facilitating decision-making in crop and disease management. Furthermore, integrating technology into crop health monitoring using GPS, GIS, UAVs, and sensors makes it possible to optimize nutrient and water usage, thereby enabling more efficient farming practices. Machine learning algorithms such as Linear Regression, Gradient Boosting, Artificial Neural Network (ANN), K-nearest neighbors (KNN), and others are assessed on a dataset for crop yield prediction, enhancing decision-making in precision agriculture. Despite the significant advancements in PA technologies, barriers to widespread adoption remain, primarily due to economic constraints, technical complexity, and limited educational outreach. This paper concludes by highlighting potential research directions that emphasize the need for cost-effective solutions and improved accessibility. This survey points towards considerable implications for a sustainable food production system, conservation of the environment, and potential progress regarding agricultural development, while it furnishes important information that can guide researchers, practitioners, and policymakers in the effort to support precision agriculture.

Index Terms—Precision Agriculture, VRT, Hydroponics, Crop Health

I. INTRODUCTION

Precision agriculture is also identified as precision farming, and this refers to the new approach in farm management by using advanced technologies in observing, measuring, and responding to the differences within intra-field conditions to be more precise. Its main purpose is the delivery of precise treatment to crops and soil when they happen to be perfect in health and productivity based on some specific data gathered in real-time. Contrasted to such traditional practices wherein water, fertilizers, and pesticides are put in the whole area of the field, precision agriculture means that such inputs are tailored to every particular area within the field. The GPS and GIS technologies are implemented at the detailed level to map all the fields with sensors, drones, or satellites providing much-needed information about soil moisture, nutrient status, crop health, and environmental conditions. This method of the site helps ensure the optimum usage of resources by enhancing efficiency in removing the waste part that causes harm to the environment, thus increasing crop yields. Generally, precision

farming brings a scientific and data-based approach into an activity that basically became more controlled and measurable for farming.

Precision agriculture becomes inevitable because of the global challenges encompassing population growth, climatic changes, and resource scarcity. With the increasing global population, there is a high demand for food, thus the more that farmers need to produce in a bid to maintain a low environmental impact. Precision farming is an imperative solution because of promotes better control of inputs on farms and thus avoids excessive water, fertilizers, and pesticide use; which reduces costs while reducing environmental damage. This approach is technology-driven and empowers farmers with better decision-making skills, which leads to higher-quality crops and more sustainable farming practices.

Agriculture 4.0 provides a growing number of tools and their applications, improving crop management through the use of high technology such as drones and sensors for the early information of crop distress due to lack of nutrients or pests and diseases, so that action can be taken timely. With Variable Rate Technology (VRT), water, fertilizers, and pesticides can be applied more effectively by applying only as much of these inputs to the targeted areas. Also, precision agriculture applies artificial intelligence and machine learning in forecasting the output of a particular crop, advising the best time for planting the crops and preventing losses after harvesting which helps maximize the farmers' returns while conserving the environment by eliminating excess chemical discharge, supporting water saving strategies and improving the quality of soil. In the future, growth in machine learning, automation, and big data AI models will allow farmers to use advanced management techniques and improve the management of crop yield and disease prediction. It is also expected that the use of mechanized machines during crop farming such as tractors and drones will help in enhancing the ease of carrying out the processes and ensuring that there is a fair return in terms of output without raising the cost of labor too much and this enhances the ability of precision agriculture to play a major role towards sustainable food production in the world which is faced by global warming and depletion of resources.

II. STATUS OF PRECISION AGRICULTURE

Precision agriculture has revolutionized the modern practices of farming by applying emerging technologies such

as remote sensing, data analysis, and geolocation systems to address the spatial and temporal variability in different agricultural fields. According to Jiang et al. [1], the United States, together with the European countries, are leaders in the exploitation of UAVs and satellite imagery in the monitoring of crop health and resource usage improvement. Such methods make site-specific management possible. Through these management techniques, application of such resources that are important like fertilizer, water, and pesticides are applied exactly where they are needed. For instance, according to Roy et al. [2], greater use of machine learning algorithms would be able to predict agricultural yields or disease outbreaks, furthering better decisions, but lessening environmental impacts. For example, Weddell et al. [3] describe how variable application of nitrogen has become an important strategy for maximizing production in the United States Pacific Northwest amid the challenges due to climate and topography that lie ahead. Adoption has been retarded, however by high implementation costs, technical know-how, and erratic data infrastructure, particularly in developing regions across the world.

In India, the field of precision agriculture remains in its early developmental phase, primarily attributable to the segmented characteristics of land ownership and the substantial expenses associated with technology. However, recent progress and governmental efforts, exemplified by the Digital Agriculture Mission, seek to incorporate IoT sensors and data analytics into conventional agricultural practices in order to tackle region-specific issues such as unpredictable monsoon patterns and suboptimal productivity levels. The increased use of remote sensing and geospatial technologies forms an essential aspect underscored by Bagwan et al. [4] to enhance management, as well as monitor agricultural crops in real time. Growth in interest can also be seen in the direction of decision support systems whereby Blessy et al. [5] argue that there is an application of machine learning for providing tailor-made crop and fertilizer recommendations to farmers by ensuring maximum yields with the minimum ecological damage. Nonetheless, considerable obstacles persist, such as the elevated expense of precision instruments, insufficient infrastructure, and a limited understanding among smallholder farmers, who continue to depend on conventional techniques.

III. TOOLS AND TECHNIQUES IN PRECISION AGRICULTURE

Figure 1 presents a detailed view of the integration of ML applications in precision agriculture, in which modern agricultural practices like hydroponics, variable rate technology, and supplementary methods significantly contribute to augmenting agricultural production. Precision agriculture operates through these technologies to monitor and control farming activities. Through ML, various applications are applied to further refine these practices. Yield prediction helps predict the output of crops in the future, crop recommendation systems provide data-driven advice to a farmer about the optimal planting time, disease prediction identifies early warnings about plant ill-

nesses, and price estimation predicts market trends by relating these to future income. Thus, such applications of ML will look to increase efficiency, make agriculture more sustainable and profitable, and positively assist farmers in decision-making and overall outcomes.

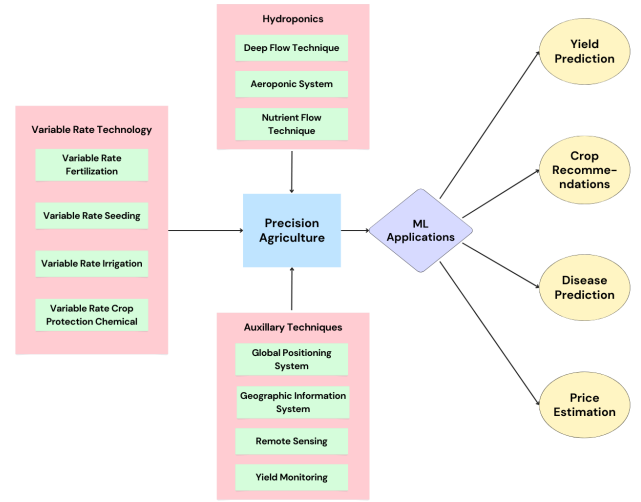


Fig. 1. ML Applications in Precision Agriculture

A. Variable-Rate Technology

Variable Rate Technology (VRT) and Variable Rate Application (VRA) represent the nucleus of modern precision agriculture. Precision agriculture is only a management concept of farming that covers achieving optimum returns on input usage while at the same time conserving resources. Precision agriculture, as such, attempts to define a decision-support system for the whole agricultural industry, and VRT and VRA represent the core of that objective. They would solve the very old problem of field variability for agriculture by giving farmers an opportunity to achieve maximum efficiency and higher yields.

That is, VRT is a system that makes machinery and equipment associated with farming work at varying rates. In simple words, it means that the rate of application of an input (such as fertilizer, seed, or pesticides) varies across a field according to the requirement of the crop at a specific location. It differs from a traditional 'blanket' approach of applying the same amount of inputs across the entire field.

VRT is accompanied by VRA or Variable Rate Application of inputs. VRA refers to the application of different types and quantities of agricultural inputs according to the needs of a specific area. VRA is based on the premise that no portion of a given field can be homogeneous; thus, even within a single field, different sections could be considered to have different types of soil, topography, characteristics of the crop type, etc., that may change their requirements for inputs. Achieved through GPS and GIS technology to map the field, sensors for measuring variation in the field, and equipment capable of changing the rate of application on the fly, the

VRT and VRA can help farmers apply accurately the right type and amount of inputs to the right place at the right time and enhance their efficiency and productivity.

The following are the 2 approaches used in VRT:

- 1) Map-based variable rate technology [6] uses pre-created prescription maps or scripts. The maps are downloaded to farm management systems or directly to agricultural machinery that supports VRT, and then the apparatus is guided in applying inputs at variable rates based on specific GPS coordinates. The map-based variable input technology further allows highly targeted applications by integrating information from multiple sources.
- 2) Variable rate technology based on sensor-based [6] variable rate applies more importance to real-time data collection for determining application rates. This technology is particularly more effective in irrigation use especially in center pivot systems, allowing the farmer to respond quickly to prevailing soil moisture, air temperature, and many other field conditions.

There are many applications of VRT in day-to-day life. Some of them are as follows:

- 1) **Variable Rate Fertilization (VRF):** Variable rate fertilization, also simply called variable rate nutrient application (VRNA) [7], applies different rates and types of fertilizers to distinctly different soil regions within a field using sensors or a pre-set field map. Because the amounts of plant-available soil nutrients, including nitrate-nitrogen, phosphorus, potassium, and sulfate-sulfur, can fluctuate substantially in regions having other soils and topography, it may be prudent to modify fertilizer types and rates on different fields for enhanced overall crop productivity. Using VRF, various types of fertilizers, including liquid, granular, gaseous, powdered, and even dung fertilizers, can be applied.
- 2) **Variable Rate Seeding (VRS):** Variable rate seeding [8] will increase crop yield with a very significant reduction in seed utilization for crop production while its precision and efficiency, based on parameters such as requirements of soil nutrients, availability of light, and water supply, adjust the seeding rate. VRS allows farmers to spatially optimize seed inputs by coordinating plant populations with fruitful regions in the field. It can also be done by a sensor-based variable rate application or satellite photography to create a prescription data map.
- 3) **Variable Rate Irrigation (VRI):** A central pivot irrigation system uses VRI [8] to help supply water to the field as efficiently as possible. An uneven field results from localized changes in the type of soil or topography. When a field is uniformly watered, certain areas may become overwatered while others may remain

unwatered. A VRI may provide a simple automatic method to avoid irrigating undesirable field sections, such as highways, streams, marshes, and other unplanted regions within a pivot.

4) **Variable Rate Crop Protection Chemical (VRC):**

Variable rate crop protection chemicals-VRC-or variable rate weeding, in which herbicide, pesticide, or insecticide application varies with location, area, and soil conditions. This method considers important factors such as the variation in infestation and weed density while spraying herbicides post-crop cultivation. Thus, weeds might be controlled more successfully while environmental risks are reduced because there are no applications of underdoses or overdoses. The method uses a sprayer and a variable rate control system for its herbicide application.

B. Hydroponics

Hydroponics is one of the new concepts that is generally referred to as growing crops in nutrients instead of soil, thus eliminating the issue of land and soil quality for farming. Water, for instance, is used in farming through a process referred to as 'hydroponics' in Greek meaning working in water [9]. This is because cultivating plants in hydroponics does not need soils, but water with high nutrient content is provided to the crops. New technologies allow monitoring of the growing environment where hydroponics is undertaken such as the pH level, temperature, and the concentration of nutrients in the growing medium which increases the growth of the plants while the risk of diseases and pests that would be harmful to the crops is greatly reduced. Economical use of Water is a key consideration when designing hydroponic systems since such systems allow recycling of water, thus they are useful in water-deficit areas. Such systems will also allow all-weather farming and can be combined with other practices such as vertical farming in order to save on land. Hydroponics is a fast-evolving technology that is easy to scale up for cultivating plants for edible use and specializes in fatty vegetables and fruit plants.

Several hydroponic techniques are widely used, each with unique mechanisms for nutrient delivery:

- 1) **Deep Flow Technique (DFT):** In DFT, the nutrient solution flows 2-3 cm deep through 10 cm diameter PVC pipes [10]. Plastic pots containing planting materials have their bottom part touching the nutrition solution flowing in pipes. Potted plants are arranged either in one plane or in zigzag shape.
- 2) **Nutrient Flow Technique (NFT):** NFT has a thin film of nutrient solution trickling down in narrow channels [9] such that the plant roots are allowed to obtain a fair portion of the nutrients. It's more efficient in its construction and has less water intake, though it's never

really well explored in aquaponics.

- 3) **Aeroponic System:** This method suspends the roots of plants in the air and sprays a nutrient-water mixture [9]. Since oxygen is constantly present, the speed of nutrient uptake increases very much, thereby making it one of the most efficient methods of plant growth.

With its many advantages, hydroponics is among the most productive and environment-friendly methods of farming. Through this method, crops can be cultivated even in poorly fertile lands or in lands with high contamination levels, thus saving fertile land and minimizing the amount of work since the entire mechanical work of tilling, cultivation, and even irrigation is eliminated. It maximizes yield potential and hence is economic in regions with the densest populations or upon costly lands. Fortunately, water and nutrients are delivered with high precision. This reduces wastage, thus preventing possible chemical contamination. Besides, hydroponically-grown plants are much less exposed to soil-borne pathogens than soil-grown crops, ensuring much healthier plants. Under hydroponics, crop growth is free from any hindrance by external environmental factors, leading to healthier plants with maximum yields. Hydroponic-produced crops are always tastier, more nutritious, and easier to harvest, having been protected from ultraviolet rays [10] or pest predators—the benefit of growing hydroponically. Hydroponics allows local cultivation even for out-of-season vegetables, which might complement local food systems and reduce transport carbon emissions.

C. Auxillary Techniques

- 1) **Global Positioning System (GPS):** Global Positioning System, or GPS technology, is applied widely in precision agriculture for enhanced field mapping, machine guidance, and improved efficiency. With GPS, the farmer can ensure placement positions, which are very important in optimizing supplies of fertilizers and pesticides by variable rate application. It saves costs and diminishes environmental impact. Maličević et al. [11] reviewed how farm mechanization systems based on GPS can increase crop yields and save 10% of input costs by avoiding overlap and optimizing equipment routes. The collection of data via GPS is also important in aiding better decision-making, which goes down well with more sustainable practices for agriculture.
- 2) **Geographic Information Systems (GIS):** Geographic Information Systems (GIS) are essential for managing and analyzing spatial data in precision agriculture. GIS allows farmers to layer data from different sources—such as soil composition, weather patterns, and yield statistics—to create detailed field maps that support decision-making. For example, Kavitha et al. [12] described how GIS-based mapping indicated that

wheat production improved by up to 15% by improving irrigations and resource allocation in Australia. GIS tools enhance farmers' ability to predict crop growth patterns, optimize inputs like water and fertilizer, and ultimately boost yield efficiency while conserving resources.

- 3) **Remote Sensing:** These technologies allow real-time monitoring of crop health and soil conditions using satellite and drone images. Non-invasive methods can detect early signs of stress, usually caused by nutrient deficiencies or diseases, allowing for early intervention. Bikkulatoova et al. [13] conducted research in Brazil that showed soybean farmers could reduce crop losses by up to 20% by using remote sensing technologies for early nutrient stress detection. This collaboration between remote sensing, GPS, and GIS can provide an extensive overview of field conditions that would permit more efficient data-informed agricultural management.
- 4) **Yield Monitoring:** Yield monitoring technology employs sensors in combines that measure crop volume extracted from various parts of a field, hence providing essential information for yield variability analysis. In the meantime, utilization of the technology helps farmers to identify spots that have high and low yields, thus allowing them to adjust agricultural inputs. This process minimizes waste, increases crop yield, and leads to better financial outcomes for farmers.

IV. MACHINE LEARNING TECHNIQUES

A. Yield Prediction Methods

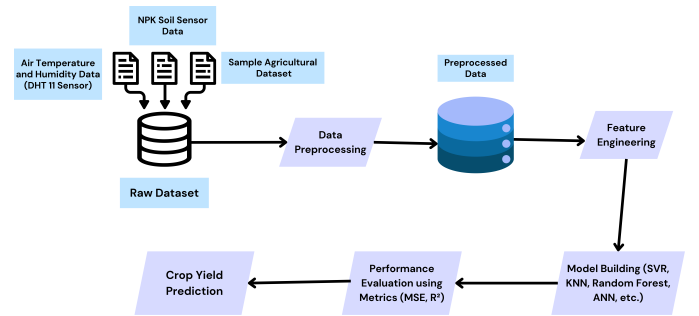


Fig. 2. Yield Prediction Workflow with ML Models

Figure 2 illustrates the workflow for predicting agricultural yield using machine learning models. Raw data, including NPK soil sensor readings, environmental data like temperature and humidity, and a sample agricultural dataset, undergo preprocessing and feature engineering to prepare it for model training. Various models such as SVR, KNN, Random Forest, ANN, etc. were employed and their suitability to predicting crop yield was assessed using performance measuring metrics like MSE and R^2 . The performance summaries and accuracy measures from these models are given in Table I.

The sample dataset [14] gives information on agricultural and environmental parameters that parameterize crop yield. It contains the columns inland rainfall in mm, fertilizer application, temperature, and soil nutrient status for nitrogen (N), phosphorus (P), and potassium (K). Each row represents a set of conditions for which these factors are measured together with yield in terms of quantity per acre for the crop concerned. This kind of information could, therefore, be used to study the dependence relationships between the variables and crop production, which could lead to the development of crop yield prediction models treating precision farming.

The chosen machine learning models effectively deal with the intricacies associated with crop yield prediction by exploiting their respective strengths in handling different types of data and relationships. These techniques are able to strike an effective balance between simplicity and complexity, thereby enabling them to capture both linear and nonlinear interactions among the key agricultural variables, such as rainfall, temperature, and soil nutrients. Their strength, ability to make effective generalizations, and low overfitting propensity make them especially appropriate for precision agriculture, in which precise predictions are of prime importance to enhance agricultural yield. Impact studies tend to orchestrate successful management and monitoring of crop health, thus ensuring the sustainable and optimal use of agricultural practices, following valid predictions.

Comparisons between models based upon the yield estimations for crops were done by utilizing MSE and R^2 scores. It is the average distance between the actual and estimated values squared that is a measure of utmost importance for estimating accuracy in agricultural datasets where that is a primary driver for decision-making. The converse modeling ability to explain input relationships in agriculture and the corresponding yield is another measure of the fitting. The R^2 score can be a good example of measuring how well models explained the percentage variation accounted for by a model. Together, these measures tell us how well the models generalize away from crop yield estimation, thus facilitating informed agricultural planning.

TABLE I
COMPARISON OF ML TECHNIQUES USING MSE AND R^2

ML Technique	Mean Squared Error	R^2 Score
Artificial Neural Network (MLP Regressor)	0.6336	0.8665
Random Forest	0.5599	0.8820
Linear Regression	0.4874	0.8973
Support Vector Regression	0.4841	0.8980
Gradient Boosting	0.4461	0.9060
K-Nearest Neighbours	0.4253	0.9104

- 1) **Artificial Neural Network (MLP Regressor):** The MLP Regressor is a feed-forward artificial neural

network capturing non-linear relationships by means of backpropagation. It worked very infirmly, with the Mean Squared Error being equal to 0.6336 and R^2 equal to 0.8665. In practice, despite being theoretically quite powerful, its tendency to overfit, together with a lack of data related to it, renders it the weakest performer among the tested models.

- 2) **Random Forest:** A step better than this was the ensemble of decision trees: Random Forest, which exhibited an MSE of 0.5599 and an R^2 score of 0.8820. Random Forest calmed overfitting and equated conduct over both linear and non-linear relations, as opposed to MLP. Nevertheless, all this still lagged far behind the predictive accuracy of simpler models, partially because it was incompletely able to capture the underlying structure of data.
- 3) **Linear Regression:** With an MSE of 0.4874 and an R^2 score of 0.8973, Linear Regression performed better than Random Forest. It managed to avoid most of the overfitting issues faced by more complex models such as Random Forest as it was able to capture linear patterns easily seen in the dataset. It was easy to use and efficient making it one of the better methods used in our study.
- 4) **Support Vector Regression (SVR):** Support Vector Regression, using the RBF kernel in order to capture the non-linear patterns, was a close second to Linear Regression, with an MSE of 0.4841 and an R^2 score of 0.8980. SVR was flexible in capturing the overall direction but not too efficient for the refinement aspect, as the ensemble-based model, Gradient Boosting did better.
- 5) **Gradient Boosting:** The Gradient Boosting was better, with an MSE of 0.4461 and an R^2 score of 0.9060, much better than the previous SVR model. The iterative refinements it enforced in the model decreased both bias and variance and resulted in a very accurate model for representing both linear and non-linear relationships.
- 6) **K-Nearest Neighbours (KNN):** The best-performing model was KNN; it achieved the lowest MSE of 0.4253 and had the highest R^2 score at 0.9104. The instance-based learning strategy of KNN captured the local feature similarities, like rainfall and temperature, thus highly suited for this data set and performing better than more complex models like Gradient Boosting.

B. Disease Prediction and Prevention

Indeed, the spread of pests and plant pathogens greatly hampers agricultural production, thus necessitating intervention through dangerous chemical agents more frequently than is liked. Nonetheless, recent developments

in Artificial Intelligence, Deep Learning, and the Internet of Things have enabled the development of sustainable approaches toward disease and pest detection early on, which could eventually make a tremendous difference in agricultural productivity. For example, for disease infestation detection in corn foliage, the highly efficient DenseNet framework suggested by Waheed et al. [15] may be used to outperform the other networks, obtaining a throughput rate of 98.0% with models such as EfficientNet and XceptionNet. Moreover, a Semi-supervised Convolutional Neural Network combined with a Graph Convolutional Network [16] is proposed and reported to achieve an impressive accuracy of 97.80% in weed identification. Kerkech et al. [17] identified vine diseases based on image data collected by UAV with an accuracy of up to 92% for grapevines. Garcia et al. [18] suggested a hybrid model concerning the fusion of Machine Learning approaches with Deep Learning methods to classify crops against weeds in Greece, obtaining an impressive F1 score of 99.29% with the use of Convolutional Neural Networks in combination with Support Vector Machine classifiers. The integration of high technologies, including Variable Rate Technology, Global Positioning Systems, and Geographic Information Systems, has greatly assisted agricultural experts in enhancing their productivity while reducing crop losses; yet chemical agents misused in agriculture remain an enduring problem. The majority of Internet of Things frameworks provide extended supervision to the integrated environment, which could prove vital in pest control; through predictive analytics, the stakeholders are empowered to use environmentally sound preventive strategies.

V. LIMITATIONS AND CHALLENGES

Precision agriculture is a newly emerging technology area that requires many challenges to be crossed over to achieve higher acceptance rates in countries where the illiteracy level among agricultural practitioners is very low. It has thus built a vast gulf between the farmers and the general advancements in modern technologies such that most farmers cannot access it. A large number of farmers tend to be indifferent towards modernizing their traditional agricultural practices and will rarely show any significant interest in developing digital skills towards upgrading agricultural practices. Besides, many agricultural areas fall in rural localities whereby internet access is limited to a high extent or, in some cases, absolutely unavailable. This also brings serious challenges for the Internet of Things (IoT) architecture and Wireless Sensor Networks (WSN) because they both need cloud services for real-time data storing and analytical processing. More critically, however, the initial investment to set up precision agriculture infrastructures, comprising both hardware and software, would be quite high. The increased costs deter small to medium-scale farmers from joining precision agriculture programs.

VI. CONCLUSION AND FUTURE FRONTIERS

Precision agriculture should be focused on most sophisticated technologies for overcoming the major challenges in present-day farming practices. These challenges include better crop identification and yield estimation under fragmented agricultural holdings, which can be achieved by the advent of satellite imagery together with normal sensor data and advanced methods of image processing: thus distinguishing various crops and assessing yield to the stipulated accuracy. Prediction models shall also have to be improved under the conditions of extreme meteorological phenomena to allow farmers prepare in advance. In this way, predictions can be complemented with more accurate and localized climate-related information so as to better equip farmers for proper use of predictive analytics in improving their capacity to respond to uncertainties brought about by climate change and extreme weather conditions.

The advancements in sensor technology, particularly multispectral imaging, will greatly aid in the identification of crop stress resulting from nutrient deficiencies, pests, and diseases. Interventions may then take place at an early stage, thereby minimizing yield losses keeping crop health in mind. Automation of land evaluation processes through the application of GIS methodology will, therefore, enhance the analysis of soil and terrain. This will lead to the efficient planning of land use and better utilization of such land. Precision agriculture is fast evolving to accommodate the small farmer and will require more economic and scalable tools, while efficient irrigation management paired with groundwater resource governance will be founded on water management technologies, namely remote sensing and hydrological modelling. All innovations will empower farmers to optimize resource use sustainably and manage agricultural practices.

REFERENCES

- [1] Y. Jiang, "Precision agriculture," in *Precision Agriculture*, 2023. [Online]. Available: <https://doi.org/10.1016/b978-0-12-822974-3.00261-5>.
- [2] A. Deb, S. Roy, S. Pyne, P. Saha, and A. Das, "Precision agriculture: Analysing the use of advanced technologies, data analytics, and remote sensing for site-specific crop management and resource optimization," in *Chapter*, 2024, pp. 53-75. [Online]. Available: <https://doi.org/10.58532/v3bcag23ch5>.
- [3] B. J. Weddell, "Precision agriculture," *International Journal of Computing Algorithm*, vol. 12, no. 1, 2023. [Online]. Available: <https://doi.org/10.20894/ijcoa.101.012.001.004>.
- [4] A. Wasim, B. Ayub, and A. Bagwan, "Revolutionizing agriculture: Geospatial technologies and precision farming in India," in *Advances in Geographical and Environmental Sciences*, Springer, 2024, pp. 43-59. [Online]. Available: https://doi.org/10.1007/978-981-97-0341-8_3.
- [5] S. S., and Blessy, "Precision agriculture decision support using machine learning," *Indian Scientific Journal of Research in Engineering and Management*, 2024. [Online]. Available: <https://doi.org/10.55041/ijrsrem29762>.
- [6] P. Naithani, "Variable rate technology in Indian agriculture scenario and solutions ahead," *ResearchGate*, 2023. [Online]. Available: https://www.researchgate.net/publication/373328054_Variable_Rate_Technology_in_Indian_Agriculture_Scenario_and_Solutions_Ahead.

- [7] C. Tom, J. Edet, T. Erokare, M. Umunna, A. Igbozulike, and N. Udenze, "Precision agriculture for maximizing crop production," *IRE Journals*, 2024. [Online]. Available: <https://www.irejournals.com/formatedpaper/1706156.pdf>.
- [8] S. Sharma, "Precision agriculture: Reviewing the advancements technologies and applications in precision agriculture for improved crop productivity and resource management," *Reviews in Food and Agriculture*, vol. 4, no. 2, pp. 45-49, 2023. [Online]. Available: <https://doi.org/10.26480/rfna.02.2023.41.45>.
- [9] E. G. Ikrang, P. O. Ehiomogue, and U. I. Udoumoh, "Hydroponics in precision agriculture – A review," *Annals of the Faculty of Engineering Hunedoara*, vol. 20, no. 2, pp. 143-148, 2022. [Online]. Available: <https://annals.fih.upt.ro/pdf-full/2022/ANNALS-2022-2-23.pdf>.
- [10] A. Swain, S. Chatterjee, and M. Vishwanath, "Hydroponics in vegetable crops: A review," *The Pharma Innovation Journal*, vol. 10, no. 6, pp. 629-634, 2021. [Online]. Available: https://www.researchgate.net/profile/Amit-Biswas-19/publication/357825388_Hydroponics_in_vegetable_crops_A_review/links/61e1149a5779d35951a7be73/Hydroponics-in-vegetable-crops-A-review.pdf.
- [11] Z. Maličević, Ž. Lakić, and M. Jugović, "Advantages and possibilities of application of precise systems in agricultural production," *AGRO-KNOWLEDGE JOURNAL*, vol. 24, no. 2, 2023. [Online]. Available: <https://doi.org/10.7251/AGREN2302043M>.
- [12] M. Kavitha, R. Srinivasan, and K. Ramadass, "GIS systems for precision agriculture and site-specific farming," in *Proceedings Title*, 2022, pp. 75-88. [Online]. Available: <https://doi.org/10.1201/9781003185413-5>.
- [13] G. Bikbulatova, E. Kupreyeva, L. Pronina, and M. Shayakhmetov, "Using remote sensing methods in precision agriculture," in *Proceedings Title*, 2020, pp. 55-59. [Online]. Available: <https://doi.org/10.2991/ASSEHR.K.200113.138>.
- [14] Yaminh, "Crop yield prediction dataset," *Kaggle*, 2021. [Online]. Available: <https://www.kaggle.com/datasets/yaminh/crop-yield-prediction/data>.
- [15] A. Waheed, M. Goyal, D. Gupta, A. Khanna, A. E. Hassanien, and H. M. Pandey, "An optimized dense convolutional neural network model for disease recognition and classification in corn leaf," *Computers and Electronics in Agriculture*, vol. 175, 2020. [Online]. Available: <https://doi.org/10.1016/j.compag.2020.105456>.
- [16] H. Jiang, C. Zhang, Y. Qiao, Z. Zhang, W. Zhang, and C. Song, "CNN feature-based graph convolutional network for weed and crop recognition in smart farming," *Computers and Electronics in Agriculture*, vol. 174, 2020. [Online]. Available: <https://doi.org/10.1016/j.compag.2020.105450>.
- [17] M. Kerkech, A. Hafiane, and R. Canals, "Vine disease detection in UAV multispectral images using optimized image registration and deep learning segmentation approach," *Computers and Electronics in Agriculture*, vol. 174, 2020. [Online]. Available: <https://doi.org/10.1016/j.compag.2020.105446>.
- [18] B. Espejo-Garcia, N. Mylonas, L. Athanasakos, S. Fountas, and I. Vasilakoglou, "Towards weeds identification assistance through transfer learning," *Computers and Electronics in Agriculture*, vol. 171, 2020. [Online]. Available: <https://doi.org/10.1016/j.compag.2019.105306>.